GESER DUGAROV, PhD | Software Engineer | Distributed Data Systems & AI Data Infrastructure

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profiles: LinkedIn, GitHub

SUMMARY

Software engineer building petabyte-scale lakehouse and AI data infrastructure at Huawei. As part of my Huawei role, I designed and implemented a two-tier distributed cache for Lance and I am currently developing a query-adaptive ANN index targeting tens of billions of vectors. I also authored 50 merged PRs to Apache Hudi and three to Lance. Research background informs an evidence-driven approach to system design, benchmarking, and performance validation. Separately, I create independent open-source agentic software-development tooling for reproducible issue-to-PR and develop–review workflows with human-in-the-loop (HITL) handoff, monitoring, and cost telemetry.

TECHNICAL SKILLS

- **Languages**: Java, Python, SQL; Rust (working knowledge)
- **Data and retrieval**: Distributed systems, performance engineering, PostgreSQL, Apache Hudi, Lance, Apache Spark, Apache Flink, Ray, vector search and ANN indexing
- **Agent systems**: Multi-agent orchestration, tool-use and HITL workflows, state and recovery management, trajectory tracing, token/cost analytics, credential isolation

WORK EXPERIENCE

May 2023 – Present — **Software Engineer / Big Data Engineer**, **Huawei**

Develop core lakehouse and AI data infrastructure for Huawei Cloud.

- Designed and implemented a two-tier distributed cache; currently developing a query-adaptive ANN index targeting tens of billions of vectors; authored three merged Rust PRs improving memory bounds and resource lifecycle in index-build jobs.
- Apache Hudi: Simplified setup of 900+ configuration parameters with recommended defaults for common scenarios and implemented partition-level TTL.

Huawei-sponsored upstream contributions

Apache Hudi – 50 merged PRs (delivered as part of the Huawei role)

- Optimized serialization of Flink stream records in Hudi, cutting end-to-end write time by up to 26% and reducing data transferred between operators by 29–36% in TPC-H benchmarks (design doc, main changes); released in Hudi 1.0.2.
- Implemented four targeted optimizations (#12054, #12104, #12113, #12120), improving Flink stream-write throughput by 10% and reducing garbage-collection overhead by 30% in internal benchmarks; released in Hudi 1.0.1.

Feb 2022 – May 2023 — **Software Engineer / ML Engineer**, **Digital Research** (computer vision startup)

- Designed an event-driven architecture and implemented the image-processing backend in Python for a truck-monitoring system handling ~20,000 images/day; in production, the system reduced fleet idle time by 12%.
- Built a customer-facing reporting and data-visualization UI and an internal monitoring UI.

INDEPENDENT PERSONAL PROJECTS

Agentic Development Tooling – Started Apr 2026

Creator and maintainer of AI-powered developer tooling for reproducible multi-agent implementation and review, human-in-the-loop handoff, monitoring, and cost tracking.

- Workflow orchestration: Built agent-orchestrator, a Python-based runner that turns a GitHub issue into an isolated implementation workflow: it launches an implementer agent, opens a PR, invokes an independent review agent, and notifies a human when the PR is ready to merge.
- Product delivery: Shipped eleven releases from v0.1.0 to v0.10.1 between May and August 2026, adding GitHub-persisted workflow state, pause/resume controls, PostgreSQL-backed usage and cost analytics, trajectory and provenance observability, namespaced workflow labels, and hardened recovery.
- Operational scale: Recorded 6,700+ agent runs across 14 repositories since May 2026, including 2,000+ review-agent runs – equivalent to $19K+ at API list prices.

EDUCATION

PhD, Geophysics, Trofimuk Institute of Petroleum Geology and Geophysics SB RAS

MSc, Computational and Applied Mathematics, Novosibirsk State University

Recent version of CV